



Made in USA

FLOATING BALL VALVE INSTALLATION, OPERATION, MAINTENANCE MANUAL

BALON Corporation

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WARNING

BEFORE ATTEMPTING TO REPAIR A VALVE THAT HAS BEEN IN SERVICE, BLOCK THE ENDS AND SLOWLY OPEN AND CLOSE THE VALVE TO VENT ANY INTERNAL PRESSURE.

BALON BALL VALVE STORAGE

Balon Corporation ships ball valves in the open position. In order to protect valves during shipment, Balon ships valves in cardboard cartons or wooden boxes. The hand wheels on gear-operated valves are removed and secured to the valves for shipment. With the exception of Series 'LM', Series 'LS', and Series W valves, valve handles are sold and shipped separately from the valves.

Valves should always be stored in the fully open position and should be kept indoors under clean, dry conditions.

BALON BALL VALVE INSTALLATION

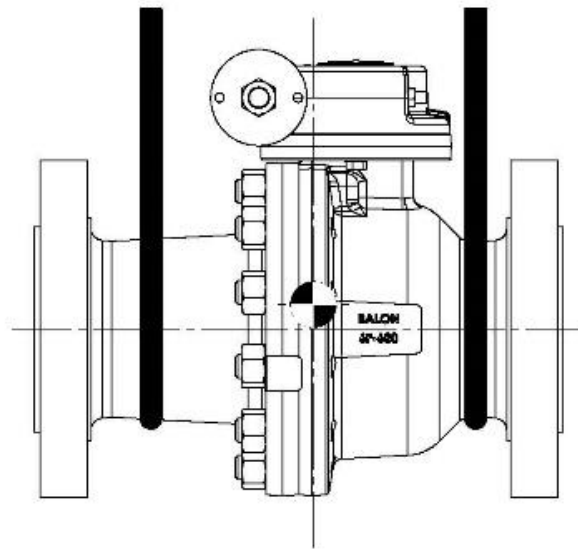
With the exception of freeze resistant uni-directional valves, Balon ball valves seal bi-directionally and may be installed in either direction.

Balon valves may be ordered as freeze resistant uni-directional valves. An arrow on the stem of a Balon uni-directional valve indicates the required flow direction. These valves are not bi-directional.

Valves should be installed in the open position. Heavy valves should be lifted into position using suitable straps.

Industry accepted make-up practice should be followed when installing flanged end valves.

Balon recommends the use of a high-quality thread compound when installing threaded end valves.



A back-up wrench should be placed on the valve component (body or end adapter) nearest to the pipe connection that is being adjusted.

Trash, such as welding slag, dirt, sand, and other foreign matter, should always be flushed from the line before operating any valve. It is recommended to fill and drain water while the valve is fully open to avoid trapping debris.

Piping systems are often pressure tested prior to their being placed in service. With valves in the open position or partially open position, valve shell testing may be done at a pressure up to 1.5 times the rated working pressure of the valves. Valves should be kept in the open position or partially open position during shell testing. Valve shell testing should never be done with the valves in the closed position.

With valves in the fully closed position, valve seat testing may be done at pressures up to 1.1 times the rated working pressure of the valves.

The pressure test medium should be eliminated from the system after testing has been completed. Water that is allowed to remain trapped inside valves can freeze. Water that freezes inside valves can damage the valves and associated pipe.

MAINTENANCE FOR BALON BALL VALVES

Balon valves require no regular maintenance to maintain proper operation and sealing. Balon stems require no adjustment or greasing for proper sealing and operation.

BALON BALL VALVE OPERATION

A ball valve is designed to be used in the fully open or fully closed position. On gear operated valves, the position indicator is located on the top of the gear box. To open the valve, rotate the handwheel counterclockwise. Using a ball valve in the partially open position can result in seat damage and, under certain conditions, damage to other valve components. Applying excessive torque in an attempt to turn a ball beyond its 90 degree stop point could damage the valve stops and misalign the ball.

A valve should always be operated within the pressure and temperature limits designated by the manufacturer.

It is good practice to operate any valve periodically in order to assure that the ball is able to move freely.

A valve should always be operated within its pressure and temperature limits. Pressure and temperature limits are based on shell components and seating material combination. See Table 1 and Table 2 for limits.

Table 1: WCB / A105 Pressure (psi) and Temperature (F°) Limits					
Pressure Class	150	300	600	900	1500
-20° - 100°	285	740	1480	2220	3705
Nylon / 212°	256	677	1354	2026	3380
Peek / 350°	215	645	1287	1932	3220

Table 2: LCC / LF2 Class 1 Pressure (psi) and Temperature (F°) Limits					
Pressure Class	150	300	600	900	1500
-50° - 100°	285	740	1480	2220	3705
Nylon / 212°	256	677	1354	2026	3380
Peek / 350°	215	645	1287	1932	3220

For pressure and temperature limits on TFE seated ball valves, consult the ratings chart on Balon's website (<http://www.Balon.com>) or contact Balon at sales@balon.com or at 405-677-3321.

For cut sheets and other information, please see:
<https://www.balon.com/downloads/CompleteCatalog.pdf>

For additional information on available repair kits or parts, contact the factory.

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Annex 1 PED Information

This is supplemental information to be used in addition to preceding information within the IOM.

Valves are considered to be Cat III to module H. Valves and nameplate are marked CE 0062.

Maximum Design Conditions are indicated on valve nameplate(s). These design conditions shall not be exceeded.

Valves are intended to be used for controlling service common to oil and gas operations. This includes, but not limited to, mixed hydrocarbons and produced water. PED group 1, fluids and/or gases. Do not use valves for any other purpose without first consulting the manufacturer.

Users of the valves are responsible for properly and safely installing valves in appropriate service applications.

Valves do not include safety relief devices. The user is responsible for providing a properly sized relieving device within piping systems to ensure piping system pressure remains below the valve's maximum allowable pressure.

Do not subject the valves to thermal or mechanical shock.

Manufacturer has not considered cyclic loading, external pressure, vibration, or decomposition of unstable fluids in the valve design. Manufacturer did not design the valves for use under external fire conditions.

To prevent operational failures, user must maintain the safe operation of the entire piping system within which the valve is installed.

Annex 1 cont. PED Information

Risk Assessment Numbers

The following information is a risk associated with the installation, maintenance, and decommissioning of valves that the Risk Assessment Number is higher than

11. Table 1

Risk	Risk Assessment Number (RAN)	Preventive Measure (PMN)
Some valves are equipped with lifting lugs. Improper lifting can cause damage to lifting lugs and hence be a potential cause for personal injury	11	3, 4, 5
Attention should be given to the environment in which any fitting, maintenance, and disassembly work is being done. Serious bodily injury, including death, may result if user fails to eliminate all sources of ignition/explosion.	15	1, 2, 3, 5, 6, 8, 9
Any fitting, maintenance, or disassembly should be performed with care.	11	1, 2, 3, 5, 6, 9, 10
Welding valves into piping systems generates heat. This can cause the valve to be hot enough to burn human skin, ignite flammable gases, or create other hazards.	15	1, 2, 3, 5, 6, 8, 9, 11
User must assess and ensure a suitable environment before installing, maintaining, or disassembling a valve.	15	1, 2, 5, 6, 7, 8, 9
User must exercise care when valves are fitted into piping systems with one end open to atmosphere and must not operate the valve from the closed to an open position if the system is under pressure.; otherwise, external leakage and / or serious injury, including death, could occur.	15	1, 2, 5, 9, 12, 13
Metallic piping systems can conduct electric current. Lightning strike, static charge, and any other sources of electric current should be considered as each is a potential source for ignition or explosion that could result in serious injury, including death.	15	1, 5, 8, 16
Ball valves must be used in the fully open or fully closed position. Never throttle through a ball valve because doing so could damage the valve or result in serious injury, including death.	11	1, 5, 6, 9, 17
User must exercise care to prevent unsafe exposure to service media. User must dispose properly of all debris or media removed from a valve and must dispose properly of the valve when it is decommissioned.	13	1, 5, 8, 9, 13, 14, 15

Annex 1 cont. PED Information

Preventive Measure

The following information outlines the preventive measures users must take to mitigate the risk. Table 2

PMN	Preventive Measure
1.	Use proper protective clothing and equipment
2.	Use proper tools
3.	Use caution and follow general safety guidelines.
4.	Use a sling or chain when lifting. Follow lifting manufacturer's guidelines when lifting. Obey all safety regulations/guidelines for lifting.
5.	Specific regulations for the equipment and standard safety regulations must be observed.
6.	See Installation, Operation, and Maintenance Manual (IOM)
7.	Balon valves are not suitable for oxygen service; a risk of combustion or explosion exists.
8.	Static charge can build-up causing an arc that could increase risk of combustion or explosion.
9.	The operator of the equipment is responsible for adhering to any and all applicable regulations.
10.	Use a high-quality thread compound when installing threaded end valves. A back-up wrench should be placed on the valve component (body or Adapter) nearest to the pipe connection that is being adjusted. Failure to do so may allow a valve joint to loosen or separate, which could cause serious injury, including death.
11.	User must follow use appropriate eye protection and follow proper welding procedures.
12.	Depressurize the piping system
13.	Provide a way to contain/capture system media.
14.	Maximum Allowable Working Pressure is displayed on the valve.
15.	Use caution before performing any work, fitting, maintenance, or disassembly
16.	Properly grounded piping system, use of insulation kits, & use of lightning protection are examples of ways to help protect against sources of ignition/explosion. Operator must adhere to all applicable regulations.
17.	Lock valve in the fully open or fully closed position.